

Table 1: Our error taxonomy with definitions, counts (Test A, Test B) and examples.

Error Type	Error Definition	Error Sub-type	Error Subtype Definition	Example
In-code Error (4, 1)	Errors within the code generated by the model.	Unresolved Loading Issue (0, 1)	Issues with loading or implementing a python library, the data, or other loading issues.	Errors loading and using geopandas and shapely [A-S115] Errors loading the data [B-41-1]
		Unresolved Processing Issue (4, 0)	Issues processing the data or prompt.	Failure to process the data, prompt, or other issues in executing the code [B-95-1]
Data Attribute Error (28, 4)	Using a different data attribute from what is in the prompt or misapplying the data attribute.	Ignores (24, 2)	Ignores the specified data attribute, could not find the right data attribute, and/or incorrectly substitutes another attribute.	Ignores the score columns and used attendance counts instead [A-S2] Ignores date column and tried to generate a date [A-S23]
		Misapplies (4, 2)	Uses the right attribute in the wrong way.	Takes the first 4 characters of the “Year” column and converts them to int) [A-S7]
Aggregation or Calculation Error (39, 1)	Using the wrong aggregation or calculation for the prompt or misapplying an aggregation or calculation.	Ignores (5, 0)	Ignores the aggregation or calculation in the prompt.	Takes an average instead of finding the correlation [A-S85]
		Misapplies (34, 1)	Performs the aggregation or calculation incorrectly.	Adds home and away team scores to China’s team [A-S27]
Grouping or Binning Error (6, 0)	Using the wrong grouping or misapplying a grouping or binning scheme.	Ignores (1, 0)	Ignores the grouping or binning specified in the prompt.	Never applied any grouping by movie [B-29-1]
		Misapplies (5, 0)	Applies the grouping or binning incorrectly.	Uses a placeholder to assign countries to continents but then fails to include other countries [A-S109]
Filter Error (134, 26)	Using the wrong filter or misapplying a correct filter.	Ignores (5, 0)	Ignores the filter in the prompt and/or applies the wrong filter.	Doesn’t apply any filter to include data for countries, so the top 10 countries includes World, IDA & IBRD Total, Low & middle income, etc [A-WB75]
		Misapplies (129, 26)	Applies the required filter incorrectly.	Filters using the string “final,” which doesn’t narrow to the final round, but also includes quarter-final and semi-final [A-S45]
Sorting Error (9, 14)	Using the wrong sorting or misapplying a sorting function.	Ignores (1, 3)	Ignores a sorting order in the prompt and/or applies the wrong sorting function.	Does not apply any sorting or reference it in the code provided [A-S28]
		Misapplies (8, 11)	Applies the required sorting incorrectly.	Sorts by sub-category in alphabetical order instead of ascending order by count [B-110-0]
Encoding Error (77, 8)	Using the wrong visual encoding or misapplying an encoding.	Ignores (39, 2)	Ignores an encoding in the prompt and/or applies the wrong encoding.	Doesn’t code separate charts for each ship mode [B-22-0]
		Misapplies (38, 6)	Apply the required encoding incorrectly.	Produces a regular bar chart when a stacked bar chart is specified [A-WB40]
Description Error (51, 1)	The underlying sentence describing the result.	Text doesn’t reflect the chart (51, 1)	Describes the chart incorrectly.	Codes a scatterplot but describes it as a dot plot [A-WB52]
Interpretation Error (13, 20)	In some responses, an additional paragraph of text follows the descriptive sentence.	Speculation (4, 3)	Makes a statement that goes beyond the available data without evidence.	Model response: “The plot reveals how different countries might prioritize the number of cylinders or focus on designs with certain acceleration levels. ” [B-16-1]
		Fabrication (9, 17)	Hallucinates - provably wrong given the provided information.	Model response: “Vehicles with 4 and 5 cylinders tend to have higher average MPG, with 5-cylinder vehicles achieving the highest average. ” [B-4-1]
Chart Design Issues (275, 386)	Code Design: Questionable or inappropriate coding for the given prompt.	Code Design (147, 1)	Questionable assumptions that may or may not lead to one of the types of errors listed above.	Model-generated code comment: # We will assume matches_df lacks attendance info and use world_cup_women_df for attendance filtering [A-S86]
	Chart Design: Issues with the chart design separate from errors, but impacting readability.	Overplotting (40, 1)	Data or labels overlap, rendering the chart unreadable.	Chart cannot be read given the overlapping axis label text [A-WB34]
		Color Choice (9, 135)	Color selection makes it hard to differentiate categories.	Different data attributes are presented using very similar colors, rendering them difficult to differentiate [B-56-0]
		Legend Design (18, 13)	Legend design, exclusion, or placement impacts chart readability.	Suppresses legend despite use of color coding in chart [B-124-0]
		Axis Setup (38, 133)	Axis set-up that hinders interpretation, comparison, or could misrepresent the data.	Uses different ranges for the same axes in small multiples [B-133-0]
		Number Format (23, 103)	Number format makes chart comprehension unnecessarily difficult.	Not using commas when presenting values in the billions in full [B-20-0]

Prompt references: A = Test A, B = Test B, S = Soccer dataset, WB = World Bank dataset. For example, [A-S115] is Test A Soccer Data Prompt 115 and [B-41-1] is Test B Prompt 41-1.